California State University, Long Beach

Mechanical and Aerospace Engineering Department

MAE 300: Engineering Instrumentation and Measurement

Experiment No. 5

Experimental Investigation of Natural Convection from a Sphere, Using Lumped System Analysis

Lab Section 19

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Abstract:

This convection heat transfer experiment aims to validate the lumped system assumptions and use that approach to calculate the heat transfer coefficient (h) of a brass and steel sphere. The convection occurring in this experiment is the heat from the brass and steel spheres transferring to the cooler room temperature air. Each sphere was heated to 120 °C by a heat gun, and their temperatures were recorded every 30 seconds by a connected thermocouple. 50 data points were collected over the course of 25 minutes. Students linearized the data to calculate the Rayleigh and Nusselt numbers to find the heat transfer coefficient. Students also calculated and compared the theoretical heat transfer coefficient to the experimental one. For brass, the calculated h was 11.314 and the theoretical h was 16.527. For steel, the calculated h was 11.832 and the theoretical h was 16.914. The values were expected as the anticipated error was 30%, and their values were within the same order of magnitude.

Introduction:

Heat transfer is defined as the transfer of energy, as heat, from one location to another. Heat transfer can occur through convection (movement of a fluid), radiation (through electromagnetic waves), or conduction (from molecule to molecule). Understanding heat transfer is invaluable in the engineering design process of rockets, engines, refrigerators, and many other mechanical systems. Heat transfer also plays a big role in the analysis of systems for energy efficiency, safety, and optimization of such systems. Heat transfer of an object can be influenced by a variety of factors, such as internal heat resistance, external heat flow, and environmental factors and shifts. The theory of heat transfer allows for the development of engineering estimates of the rates of heat transfer for a variety of different practical systems. However, this is often an idealization and not always an accurate representation of reality. It is for this reason that methods such as the lumped capacitance method, that assume certain values as being negligible, are significant, especially the validation of such methods. Values such as the heat transfer convection coefficient, that represents the rate of heat transfer between an object's surface per unit of fluid area, can be used to provide valuable insight into how a system transfers energy through heat. In addition, the convection and radiation heat transfer values quantify the amount of heat transferred out of an object through each of these different methods. These values can be calculated through experiments, as well as theoretically via empirical engineering values.

Background & Theory:

The objective of this experiment is to determine the convective heat transfer coefficient (h) of a heated sphere as it cools in air. This is analyzed using the lumped system approach, which assumes that the temperature within the sphere is uniform at any given time. This assumption is valid when internal heat conduction is much faster than heat loss at the surface, meaning temperature gradients inside the sphere are negligible.

The total heat transfer from the sphere to the surroundings consists of both convection and radiation, as shown in Eq. (3). Convective heat transfer is described by Eq. (1), where the heat transfer coefficient (h), surface area (A), sphere temperature (T), and ambient temperature

(T_∞) are required. Radiative heat transfer is given by Eq. (6), which depends on the emissivity (ε), Stefan–Boltzmann constant (σ), and the fourth power of the temperature difference. In this experiment, radiation is relatively small compared to convection, so it is often neglected, and convection becomes the dominant mode of heat transfer.

An energy balance is applied by equating the heat leaving the sphere (Eq. (1)) with the decrease in internal energy (Eq. (2)). This leads to the combined expression shown in Eq. (3), which relates temperature change to time. By rearranging and integrating this relationship, the data can be linearized using Eq. (16), allowing a straight-line plot to be obtained. The slope of this line is then used in Eq. (4) to determine the experimental heat transfer coefficient. The required values for this calculation include the slope from the graph, as well as the density (ρ), volume (V), specific heat (C_p), and surface area (A) of the sphere.

The total heat transfer can also be expressed using the experimentally determined coefficient, as shown in Eq. (5). If radiation is considered, the convective portion of heat transfer can be isolated using Eq. (7), which subtracts the radiative contribution from the total heat transfer. This provides a more accurate representation of convection if radiation effects are significant.

The validity of the lumped system assumption is evaluated using the Biot number, defined in Eq. (8). This dimensionless parameter compares internal conduction resistance to external convection resistance. If the Biot number is less than 0.1, the temperature within the sphere can be assumed uniform, and the lumped system approach is valid.

The theoretical value of the heat transfer coefficient can also be calculated using empirical correlations. This requires several air properties, including thermal conductivity (k), kinematic viscosity (υ), and Prandtl number (Pr), which depend on temperature and are evaluated at the film temperature given by Eq. (10). In this experiment, the room temperature was approximately 21°C (about 294 K). Since this is very close to 300 K, the air properties at 300 K from the provided table were used as an approximation. The selected values were ($k = 0.0263\ W/m*K$), $\upsilon = 1.589 * 10^{-5} m^2/s$, and $Pr = 0.707$. This approximation introduces minimal error due to the small temperature difference.

Using these properties, the Rayleigh number is calculated using Eq. (9), which accounts for buoyancy-driven flow and depends on gravity (g), thermal expansion coefficient β, sphere diameter (D), temperature difference, and fluid properties. The Prandtl number is obtained directly from the table. The Nusselt number is then determined using Eq. (12), which provides a correlation for natural convection around a sphere under the given conditions. This value represents the enhancement of heat transfer due to convection compared to pure conduction.

Finally, the theoretical heat transfer coefficient is calculated using Eq. (13), which relates the Nusselt number to the thermal conductivity of air and the diameter of the sphere. The average surface temperature used in these calculations can be determined using Eq. (11). The theoretical convective heat transfer rate can then be found using Eq. (14). Additionally, uncertainty in the experimental heat transfer coefficient may be evaluated using Eq. (15), which accounts for measurement errors in temperature and logarithmic calculations.

1) $Q_{out} = Q_h = hA(T - T_\infty)$
2) $-Q_{internal} = -\rho V C_p \frac{dT}{dt}$

3) $q_{total} = A(T - T_\infty) = -\rho V C_p \frac{dT}{dt}$
4) $h_{exp} = \frac{-slope \cdot \rho V C_p}{A}$
5) $q_{total} = h_{exp} A(T_W - T_\infty)$
6) $Q_{radiation} = \sigma \varepsilon A(T_w^4 - T_\infty^4)$
7) $Q_{convection} = Q_{total} - Q_{radiation}$
8) $Biot\# = \frac{h.D}{6K_s} = \frac{h_{exp}.D}{6K_s}$
9) $Ra = \frac{g\beta D^3(T_w - T_\infty)}{\nu^2} Pr$
10) $T_f = \frac{T_w + T_\infty}{2}$
11) $T_w = \frac{T_{w1} + T_{w2}}{2}$
12) $Nu = 2 + \frac{0.589 Ra^{1/4}}{(1+(0.469/Pr)^{9/16})^{4/9}}$ $for\ Pr \geq 0.7\ and\ Ra \leq 10^{11}$
13) $h = \frac{Nu * K}{D}$
14) $Q_{theoretical\ convection} = h_{ave} A(T_w - T_\infty)$
15) $U_h = \pm \left[\left(\frac{\Delta T}{(T - T_\infty) \ln(\frac{T - T_\infty}{T_i - T_\infty})} \right)^2 \right]^{\frac{1}{2}}$
16) $y = ln(\frac{T - T_\infty}{T_i - T_\infty})$

Experimental Procedures:

Brass and stainless-steel solid spheres were hung on the apparatus as shown in Figure 1.1. The stainless-steel sphere is $D = 0.01881m$ in diameter, with density $\rho = 7460\ kg/m^3$, specific heat of $460\ J/kg * K$, thermal conductivity of $14\ W/m * K$, emissivity of 0.07, and kinematic viscosity of $0.00001589\ m^2/s$. The brass sphere has the same diameter, with density $8730\ kg/m^3$, specific heat of $376\ J/kg * K$, thermal conductivity of $119\ W/m * K$, emissivity of 0.52, and kinematic viscosity of $0.00001589\ m^2/s$. With the same diameter, the volume and surface were calculated, $3.4847E - 06\ m^3$ and $0.00111m^2$, respectively. The stainless-steel and brass spheres were heated to 120 °C with a heat gun. Then allowed them to cool down while the temperature of the spheres was recorded every 30 seconds for 1500 seconds. This data was used to calculate experimental and theoretical convection heat transfer coefficient via Eq. (13).

Figure 1.1: Sphere on Stand



Raw data:

Figure 1.2: Brass Temperature Data

Time (s)	Temp [C]	$ln(\frac{T - T_\infty}{T_i - T_\infty})$
0	120.1	0
30	109.8	-0.110
60	105.8	-0.156
90	94.6	-0.297
120	93.2	-0.317
150	91	-0.348
180	87.6	-0.397
210	84.4	-0.447
240	82.5	-0.477
270	79.9	-0.520
300	77.5	-0.562
330	75.1	-0.605
360	72.3	-0.658
390	70.1	-0.702
420	68.3	-0.740

450	67.3	-0.761
480	65.3	-0.805
510	62.8	-0.863
540	60.2	-0.927
570	59.7	-0.940
600	56.8	-1.018
630	56.3	-1.032
660	55.1	-1.067
690	52.9	-1.134
720	52.2	-1.156
750	51.1	-1.192
780	50.3	-1.219
810	49.7	-1.239
840	47.1	-1.334
870	48.2	-1.293
900	46.5	-1.357
930	45.5	-1.397
960	45.3	-1.406
990	41.7	-1.566
1020	41	-1.600
1050	40.2	-1.641
1080	39.5	-1.678
1110	39	-1.706
1140	38.8	-1.717
1170	37.6	-1.787
1200	37.6	-1.787
1230	37.1	-1.817
1260	36.6	-1.849
1290	35	-1.957
1320	34.8	-1.971
1350	34	-2.031
1380	33.2	-2.095
1410	33.7	-2.055
1440	33.2	-2.095
1470	32.6	-2.145
1500	31.6	-2.235

Figure 1.3: Steel Temperature Data

Time (s)	Temp [C]	$ln(\frac{T-T_\infty}{T_i-T_\infty})$
0	120	0.000
30	109.4	-0.113

60	91.1	-0.345
90	89.2	-0.373
120	85.9	-0.422
150	83.3	-0.463
180	81.5	-0.492
210	78.8	-0.538
240	76.1	-0.586
270	74.1	-0.623
300	72.1	-0.661
330	70.1	-0.701
360	68.1	-0.743
390	66.5	-0.777
420	64.9	-0.813
450	63	-0.857
480	61	-0.906
510	59.8	-0.937
540	58.2	-0.979
570	57.1	-1.009
600	55.8	-1.046
630	54.6	-1.081
660	53.5	-1.114
690	51.8	-1.168
720	50.8	-1.201
750	50.1	-1.224
780	49	-1.263
810	48	-1.299
840	47.3	-1.326
870	46.6	-1.353
900	46.1	-1.372
930	44.7	-1.430
960	44.2	-1.451
990	43.5	-1.482
1020	42.5	-1.527
1050	42.3	-1.536
1080	41.4	-1.580
1110	41	-1.599
1140	40.5	-1.625
1170	40	-1.651
1200	39.1	-1.699
1230	39	-1.705
1260	38.5	-1.733
1290	38.3	-1.744
1320	37.7	-1.780

1350	36.9	-1.829
1380	36.7	-1.841
1410	36.2	-1.874
1440	35.8	-1.900
1470	35.6	-1.914
1500	35.2	-1.942

Sample Calculations and Results:

Figure 1.4: Brass Linearization Graph

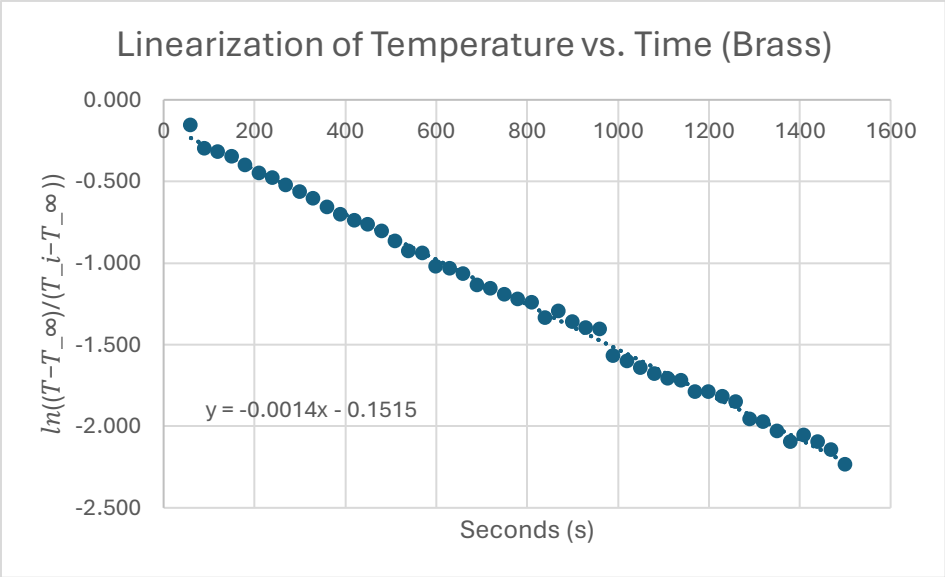
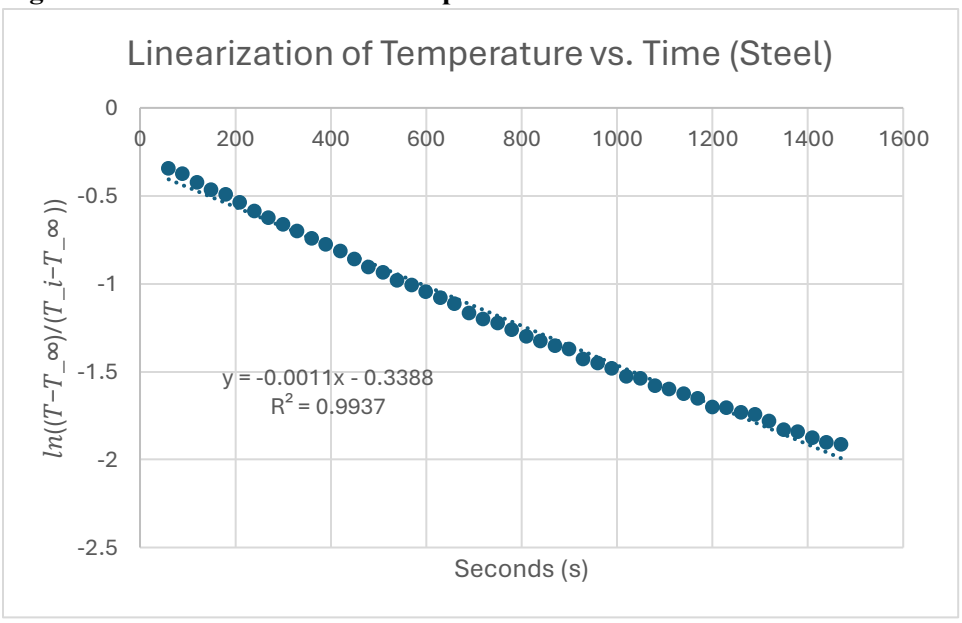


Figure 1.5: Steel Linearization Graph



Brass Sample Calculations:

Slope Time Interval of 60 s to 1500 s:

$$a_0 = \frac{\Sigma x_i \Sigma x_i y_i - \Sigma y_i \Sigma x_i^2}{(\Sigma x_i)^2 - n(\Sigma x_i)^2} = \frac{(38220 * -59034.2) - (-60.1 * 38631600)}{(38220)^2 - 48 * (38220)^2} = -0.1515$$

$$a_1 = \frac{\Sigma x_i \Sigma y_i - n\Sigma x_i y_i}{(\Sigma x_i)^2 - n\Sigma x_i^2} = \frac{(38220 * -60.1) - (48 * -59034.2)}{(38220)^2 - (48 * 38631600)} = -0.0014$$

$$y = a_0 + a_i x = -0.0014 - 0.1515\text{x} \pm 0.5$$

Uncertainty for Experimental Heat Transfer Coefficient:

$$U_h = \pm\left[\left(\frac{\Delta T}{(T_w - T_\infty)\ln(\frac{T_w - T_\infty}{T_i - T_\infty})}\right)^2\right]^{\frac{1}{2}} = \pm\left[\left(\frac{0.5}{(68.7 - 21)\ln(\frac{68.7 - 21}{120.1 - 21})}\right)^2\right]^{\frac{1}{2}} = \pm 0.0143$$

$$\Delta h_{exp} = U_h \cdot h_{exp} = (0.0143)(11.314) = \mathbf{0.1622}$$

Brass Experimental Heat Transfer Coefficient:

$$h_{exp} = \frac{-slope \cdot \rho V C_p}{A} = \frac{-0.0014(8800)(3.48E-06)(375)}{0.0011} = \mathbf{11.314 \pm 0.1622}$$

Calculation of T_w for Brass:

$$T_w = \frac{T_{w1} + T_{w2}}{2} = \frac{105.8 + 31.6}{2} = \mathbf{68.7}$$

Total Heat Transfer for Brass:

$$q_{total} = h_{exp} A(T_W - T_\infty) = (11.314)(0.0011)(68.7 - 21) = \mathbf{0.599}$$

Radiation Heat Transfer for Brass:

$$Q_{radiation} = \sigma \varepsilon A(T_w^4 - T_\infty^4) = (5.67E-0.8)(0.4)(0.0011)(68.7^4 - 21^4) = \mathbf{5.56E-04}$$

Experimental Heat Transfer by Convection:

$$Q_{convection} = Q_{total} - Q_{radiation} = 0.599 - 5.56E-04 = \mathbf{5.99E-01}$$

Brass Rayleign Number:

$$T_f = \frac{T_w + T_\infty}{2} = \frac{68.7 + 21}{2} = 44.85$$

$$\beta = \frac{1}{T_f} = \frac{1}{44.85} = 0.0223$$

$$Ra = \frac{g\beta D^3 (T_w - T_\infty)}{\nu^2} Pr = \frac{9.81(0.0223)(0.0188)^3(68.7 - 21)}{(1.5E-05)^2} 0.707 = \mathbf{217838.87}$$

Brass Nusselt Number:

$$Nu = 2 + \frac{0.589 Ra^{1/4}}{(1 + (0.469/Pr)^{9/16})^{4/9}} = 2 + \frac{0.589(217838.87)^{1/4}}{(1 + (0.469/0.707))^{9/16})^{4/9}} = \mathbf{11.81}$$

Theoretical Heat Transfer Coefficient:

$$h_{theoretical} = \frac{Nu * K}{D} = \frac{(11.81)(0.0263)}{0.0188} = \mathbf{16.527}$$

Theoretical Heat Transfer by Convection:

$$Q_{theoretical\ convection} = hA(T_w - T_\infty) = (16.527)(0.0011)(68.7 - 21) = \mathbf{0.867}$$

Biot# for Brass:

$$Biot\# = \frac{h.D}{6K_s} = \frac{h_{exp}.D}{6K_s} = \frac{(11.314)(0.0188)}{6(119)} = \mathbf{0.0003}$$

Figure 1.6: Comparison of Calculated Values

Calculated Values	Brass	Steel
$y = a_0 + a_i x$	$y = -0.0014 - 0.1515\text{x} \pm 0.5$	$y = -0.0011 - 0.3388\text{x} \pm 0.5$
h_{exp}	11.314 ± 0.1622	11.832 ± 0.2142
T_w	68.7	63.15
q_{total}	0.599	0.554
$Q_{radiation}$	$5.56E-05$	$6.93E-05$
$Q_{convection}$	0.599	0.554
Ra	217838.87	244040.06
Nu	11.81	12.097
$h_{theoretical}$	16.527	16.914
$Biot\#$	0.0003	0.00265
$Q_{theoretical\ convection}$	0.867	0.791

Discussion of results:

The temperature measurements of a steel and brass sphere were collected over a period of 25 minutes in intervals of 30, providing 50 data points. For both data sets, using (16) a linearized graph was plotted over the time interval of 60 seconds to 1500 seconds in which the slope of the data points was used to calculate the experimental heat transfer coefficient. The reason that the graph was plotted from the 60 second mark was because the temperature drop stabilized after a minute, as the initial drop in temperature would potentially skew the data. The slope and equation of the line of best fit for the graph was calculated using the method of least squares. In addition, the theoretical value for the heat transfer coefficients was calculated by finding the Rayleigh number (9) and the Nusselt number (12). The values used to calculate the Rayleigh and Nusselt numbers were taken for the temperature value being the average of the temperature values at 60 seconds and 1500 seconds. For the brass sphere this average temperature value (T_w) is 68.7 °C and 63.15 °C for the steel sphere. The calculated values for the experimental heat transfer coefficient for brass is 11.314 ± 0.1622 and 11.832 ± 0.2142 for the steel sphere. The theoretical heat transfer coefficient for brass is 16.527 and 16.914 for steel. Comparison of the theoretical and experimental values for the brass sphere proves that although there is a percentage error of around 30%, the values both fall in the same order of magnitude. This applies to the values for the steel sphere as there is the same error percentage with the values being on the same order of magnitude.

The radiation heat transfer values and the convection heat transfer values for both the brass and steel sphere were compared to determine if the radiation heat transfer is rightfully assumed to be negligible. Radiation heat transfer is the transfer of energy through electromagnetic radiation while convection heat transfer is the transfer of energy through the movement of a fluid. In this experiment forced and natural convection is taking place as there is an external device, heat gun, driving the forces for the fluid, surrounding air, to move, while the surrounding air naturally transfers the heat of the sphere into the surroundings. The calculated value for the radiation heat transfer for the brass sphere is $5.56E-05$ and the steel sphere is $6.93E-05$. The convection heat transfer value for the brass is 0.599 and the value for the steel is 0.554. Both the radiation and convection values for the brass and steel are relatively similar and have the same order of magnitude. But when comparing the values of radiation versus convection, the values of radiation are around 4 orders of magnitude lower than that of the convection values. This means that it is correct to assume that the radiation heat transfer is negligible for both the brass and steel spheres.

The calculation of Biot number is required for its verification. The lumped capacitance or lumped system analysis is used to determine the transient, or temporary, heat transfer of an object. This approach assumed that the internal thermal resistance of an object is minimal in comparison to the external flow resistance and is negligible. The temperature of the object is assumed to be uniform throughout its interior, and the temperature change of the object is considered to only be a function of time. The Biot number is the ratio of the internal resistance of the object and the external flow resistance of the surface of the object. The internal resistance is found by multiplying the experimental heat transfer coefficient with the diameter of the sphere. The external flow resistance is found by multiplying the thermal conductivity of the material by six. The lumped system method is considered valid if the corresponding Biot number for the system is less than 0.01. The Biot number for brass is 0.0003 and 0.00265 for steel. Both values are less than 0.01, therefore the lumped system method is valid for both spheres. One interesting observation can be made when comparing both values of the Biot number, as the Biot number for the brass was one order of magnitude less than the steel. As the experimental values of the heat transfer coefficient were relatively the same, the difference can be attributed to the thermal conductivity of each of the materials. As the thermal conductivity of the brass sphere was around 119 W/mK while the steel was 14 W/mK. This means that the brass is a better conductor of heat than the steel, so it would be easier for the internal temperature of the brass sphere to balance out and resemble a uniform internal heat distribution.

Another comparison that provides insight into the potential discrepancies of the lumped system approach would be the comparison of the experimental and theoretical convection heat transfer values. As stated previously, the impact of the radiation heat transfer is minimal in comparison to the convection heat transfer, and therefore it is negligible and omitted in our analysis. The theoretical value for the convection heat transfer is 0.867 for the brass and 0.791 for the steel. The experimental values for the brass are 0.599 and 0.554. The theoretical values for both materials are higher than the experimental values. This discrepancy can be attributed to heat losses from a variety of areas within the experimental procedure. For instance, the surrounding air around each of the two spheres was impacted by the number of people surrounding them during the data collection. As the main source of heat transfer was determined

to be through convection, the movement of many different bodies around the two spheres could have impacted the values recorded in the experiment. Since at the beginning of the experiment, many individuals crowded around the measurement devices to record the results, the temperature measurements would have been slightly altered from the theoretical values. Since the experimental values were lower than the theoretical values, this indicates that there was a higher anticipated heat transfer rate than what happened. As less heat was transferred this indicates that because of the presence of multiple heat-emitting bodies, the spheres couldn't transfer heat into the surroundings as predicted. Another source of error could have come from location of the two spheres being relatively close to on another that would have affected the temperature recordings. In addition, the potential inaccuracy of the equipment could also be considered as there was no accuracy test or calibration conducted on the two apparatuses.

Conclusions and Recommendations:

This experiment tested the validity of the lumped system approach in analyzing the heat transfer of a steel and brass sphere after being heated with a heat gun to a temperature of 120 °C. The lumped system approach assumes that the internal thermal resistance of an object is negligible in comparison to the external flow resistance. To test this approach's validity the experimental heat transfer coefficient was found by linearizing the temperature data of the two spheres that was collected over a 25 minute interval at 30 second intervals. The method of least squares was used to find the slope and intercept values for each of the graphs with the slope being used to calculate the experimental heat transfer coefficient. The calculated values for the experimental heat transfer coefficient for brass is 11.314 ± 0.1622 and 11.832 ± 0.2142 for the steel sphere. In addition, the theoretical heat transfer coefficients were calculated by using (13) and finding the Rayleigh and Nusselt numbers at the time (T_w), which is an average of the temperatures at the time 60 seconds and 1500 seconds. The theoretical heat transfer coefficient for brass is 16.527 and 16.914 for steel. For both materials, the percentage of errors between experimental and theoretical values was around 30%.

The Biot number was used to test the validity of the lumped system approach for this instance, as a Biot number lower than 0.01 would indicate that the lumped system approach was valid. For both the brass and steel spheres, the Biot number was lower than 0.01, with the brass Biot number being 0.0003 and the steel being 0.00265. An additional analysis of the experimental and theoretical convection heat transfer values were used to identify potential sources of heat loss in the experimental procedure. The theoretical value for the convection heat transfer is 0.867 for the brass and 0.791 for the steel. The experimental values for the brass are 0.599 and 0.554. The lower experimental values indicate potential errors within the data collection such as the proximity of the steel and brass spheres and the interference of body heat on the spheres.

All in all, this experiment emphasizes the importance of data analysis when choosing a method for calculating a value from experimental data. As an incorrect assumption would negatively affect the accuracy of calculated values such as the heat transfer coefficient. This can lead to a cascading effect of inaccuracies within the design, analysis, and engineering process.

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